

Name: _____ Mark: _____ /70

G10 Science - Class 1 Homework

1. Draw the Bohr-Rutherford Diagrams for the following elements: **[3 marks]**

a) Nitrogen (N-14)

b) Aluminum (Al-27)

c) Chlorine (Cl-35)

2. Draw the Lewis Dot Diagrams for the following elements: **[3 marks]**

a) Bromine

b) Oxygen

c) Lithium Ion (Li^+)

3. Compare (similarity) and contrast (differences):

a) Ion and Isotope **[3 marks]**

b) Ionic and Covalent **[3 marks]**

c) Cation and Anion **[3 marks]**

4. Complete the following table for the following ions: **[5 marks]**

Ion	Name	Number of Protons	Number of Neutrons	Number of Electrons
Mg^{2+}				
	Sulfide ion			
		26		23
Br^-				
	Nitride ion			

5. Draw the Lewis Dot Structure for magnesium and chlorine and show the transfer of electrons to make an ionic compound. **[4 marks]**
6. Hydrogen peroxide H_2O_2 is a molecular compound used to disinfect cuts. Why is this formula of this compound not written as HO ? **[2 marks]**
7. An unknown element X forms a compound with chlorine: XCl_2 . Predict the chemical formula of the compound that element X makes with oxygen. Justify your answer. **[2 marks]**
8. A student claims that SnCl_4 is named tin chloride. Explain why the chemical name is incorrect and provide the correct answer for SnCl_4 . **[2 marks]**

9. Classify each of the following as ionic (I) or covalent (C), then write the chemical name:
[20 marks]

Chemical Formula	Ionic/Covalent	Chemical Name
CaF ₂		
N ₂ O		
Al ₂ O ₃		
SO ₂		
Ca ₃ P ₂		
PbCl ₂		
Fe ₂ O ₃		
SnS		
Cu ₃ P ₂		
CaCrO ₄		
CuF ₂		
N ₂ O ₄		
Cu ₃ P		
PCl ₃		
MnO		
KMnO ₄		
CaS		
CoN		
SF ₆		
KClO ₃		

10. Classify each of the following as ionic (I) or covalent (C), then write the chemical formula:
[20 marks]

Chemical Name	Ionic/Covalent	Chemical Formula
Calcium chloride		
Aluminum phosphate		
Magnesium sulfide		
Lithium nitride		
Diphosphorus tetrasulfide		
Iron(II) bromide		
Manganese(IV) oxide		
Phosphorus pentafluoride		
Copper(I) sulfide		
Iron(III) nitride		
Copper(II) oxide		
Silver acetate		
Cobalt(III) oxide		
Lead(II) nitride		
Aluminum sulfide		
Carbon tetrachloride		
Beryllium hydroxide		
Sodium phosphide		
Nitrogen dioxide		
Iodine heptafluoride		